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RESEARCH NOTE

Chi-Hung Juan · Vincent Walsh

Feedback to V1: a reverse hierarchy in vision

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Abstract We examined the involvement of visual area V1 in visual detection to assess the role of feedback connections to V1 as proposed in reverse hierarchy theory (Ahissar and Hochstein 2000). In Experiment 1, signal detection was decreased in feature and conjunction detection tasks by repetitive pulse TMS (rTMS) over V1 for 500 ms after stimulus onset. In Experiment 2, rTMS was delayed to allow uninterrupted signal processing for 100 ms after visual stimulus onset. TMS for the subsequent 500 ms now disrupted detection of conjunction but not feature targets. In Experiment 3 we applied double pulse TMS at varying intervals to assess the timing of V1 involvement in these tasks. Single feature detection involved V1 only at some point between 40 and 100 ms after visual array onset; detection of targets defined by conjunctions of features involved V1 throughout the first 100 ms and also between 200 and 240 ms after visual stimulus onset. We suggest that the early effects in the conjunction task are due to repeated sampling of the visual array to extract the signal from external noise. The later effects in conjunction search are attributed to the return projections from secondary visual areas back to V1, consistent with the reverse hierarchy theory. The effects in both tasks are consistent with early and repeated iterations of feed forward and feedback loops as hypothesised in recent neurophysiological experiments (see Foxe and Simpson 2002).

Keywords Transcranial magnetic stimulation · Double pulse TMS · V1 · Reverse hierarchy theory · Visual search · Delayed onset rTMS

C.-H. Juan (✉)
Department of Psychology, Vanderbilt University,
301 Wilson Hall, Nashville, TN 37240, USA
e-mail: chi-hung.juan@vanderbilt.edu
Tel.: +1-615-3437538
Fax: +1-615-3438449

V. Walsh
Institute of Cognitive Neuroscience,
University College London,
Alexandra House, 17 Queen Sq., London, WC1N 3AR, UK

Introduction

Primary visual cortex is often seen as the first and lowest stage of a feed forward cortical visual hierarchy (see Hubel and Wiesel 1977). This view is based on factors such as small receptive fields, the apparently simple nature of response properties (bars, wavelengths and spots for example) and the timing of responses to visual input, which can be as early as 20 ms (see Bullier 2001). As a consequence it has been thought that V1 has little contribution to make to the analysis of visual arrays in which the signal to noise ratio is low, the stimuli are complex, or in which some variation of attention is said to be required to carry out the task.

This feed forward view of vision has been challenged. Anatomically, the feedback pathways from secondary visual areas to V1 provide a substrate for complex interactions in which V1 may be the vital "look up table" (Ahissar and Hochstein 2000). Physiological studies show that V1 cells may respond to complex contextual information in the visual scene (Zipser et al. 1996; Nothdurft et al. 1999), show responses that can be described as attentional (Motter 1993; Somers et al. 1999; Lamme and Spekreijse 2000), and respond at times later than the mean latencies for secondary visual areas (Bullier 2001). In dense EEG recordings, some components hitherto termed "early" have been shown to be due to the arrival of signals from parietal and frontal cortex (Foxe and Simpson 2002), and studies with transcranial magnetic stimulation (TMS) have shown back projections from V5 to V1 to be important for awareness (Cowey and Walsh 2000; Pascual-Leone and Walsh 2001; Ro et al. 2003). Psychological theories of visual scene analysis have concentrated on feed forward possibilities in which larger receptive fields with more complex response properties perform the critical analysis (e.g. Treisman 1996). A theoretical alternative is the reverse hierarchy theory (RHT) of visual perception (Ahissar and Hochstein 2000), which proposes that higher visual areas carry out a preliminary analysis of attributes but that V1 supplies a more detailed report of fine structure and spatial localisation.

In this study we tested the RHT by disrupting visual search performance with repetitive or double pulse TMS and suggest that V1 is required in both early and later stages of difficult visual detection tasks.

Materials and methods

Subjects

Eight subjects participated in Experiment 1, six in Experiment 2 and six in Experiment 3. They were aged 19–34 years, were right handed, had normal or corrected to normal vision and were naïve to the purpose of the experiment. Informed consent was obtained according to permissions granted by the Oxford Research Ethics Committee (OXREC).

Procedure

General

Feature and conjunction visual detection tasks were used in the experiments. Subjects reported the presence or absence of a target defined either by a unique colour (either a blue circle amongst red circles or a red circle amongst blues) or a unique combination of colour and orientation (a blue forward slash amongst an array of blue back slashes and red forward slashes). On a trial in which the target was present there were 11 distracters and on a target absent trail there were 12 non-targets. The search array subtended 2×2 degrees. Subjects viewed the stimuli binocularly, 57 cm from the screen. Each trial began with a 500-ms central fixation point followed by the search array and a visual mask. The mask was displayed until subjects made the present/absent response by a key press. Subjects were asked to respond accurately without any time pressure. Performance was manipulated by varying the stimulus-mask onset asynchrony (SOA) in steps of 10 ms in a staircase procedure until performance reached between 62.5% and 87.5% correct. The mean SOA in the feature task was 50 ms (range 30–100 ms) and in the conjunction task 120 ms (range 80–180 ms). Responses were subjected to signal detection analysis (Green and Swets 1966).

In each experiment there was a block of 160 feature trials and a block of 160 conjunction trials. TMS was delivered randomly on half the trials.

Magnetic stimulation

A Magstim Super-Rapid TMS machine with a 70-mm figure of eight coil was used. Stimulation intensity was set at 65% of stimulator output. The coil was centred approximately 1 cm above the inion, selected on the basis of phosphene production or scotomas induced in previous experiments. Subjects did not perceive phosphenes *during* the presentation of the visual arrays.

We have shown previously that stimulation over this region principally excites V1 (Cowey and Walsh 2000; Pascual-Leone and Walsh 2001; see also Kammer 1999) and will use this shorthand throughout. In our view there is too much emphasis on the simple matter of current spread on this issue. Inspection of the *functional* consequences of occipital TMS leaves very little room for an argument that V1 is not the site of interest being stimulated. First of all, the suggestion that phosphenes and scotomas are caused by subcortical stimulation does not explain why low intensity and high intensity TMS produce the same phosphenes at the same location with only differences in apparent brightness. The subcortical argument requires the stimulation to pass through the cortex (having no effect) whereas a cortical basis merely requires that stimulation has different effects on the first point at which induced current meets tissue (cortex). Further, low intensity TMS can produce phosphenes even though the current induced subcortically would be negligible. Second, the argument that V2 (or some would say even V3) is the site of interest is incompatible with too many effects of stimulation: there is no hypothesis that backprojections to V2 or V3 mediate awareness (Pascual-Leone and Walsh 2001); there is no doubt that the blindsight patient GY's V2 can be stimulated, but there is nevertheless no consequence of this; the timing of occipital effects with V1 does not fit V2 or V3 physiology; the size of scotomas fits V1 (also V2) but the absence of movement does not fit V3. In summary, while stimulating V2 without V1 would be very difficult, the default should be that unless shown otherwise, V1 is the most highly probably default. In the context of this paper, there is no evidence to suggest that V2 is a candidate for the effects observed.

Experiment 1: rTMS

Subjects received rTMS at 10 Hz for 500 ms at the onset of the visual stimulus in order to establish whether V1 disruption would have any effect on either of the visual search tasks.

Experiment 2: gap rTMS

Subjects received rTMS at 10 Hz for 500 ms with the onset of rTMS beginning 100 ms after the onset of the visual array. This was to test whether the first 100 ms, if left unstimulated, would be sufficient for V1 to carry out all the processing required in feature and conjunction tasks.

Experiment 3: dTMS

Subjects received double pulse TMS (dTMS) to try to establish if the role of V1 in feature search was confined to a particular range of time within the first 100 ms after stimulus onset and to test whether the role of V1 in conjunction search had any temporal specificity. In the feature search task, which Experiment 2 showed to involve V1 only in the first 100 ms, subjects received dTMS at 0 and 40 ms, 0 and 100 ms and 80 and 120 ms after the onset of the visual array. In the conjunction search task, which, as Experiment 2 showed, requires V1 involvement outside the first 100 ms (we could not say, without Experiment 3, whether the first 100 ms is also important), subjects received dTMS at the same times as in the feature task and also at 140 and 180, 200 and 240 and 260 and 300 ms after stimulus onset.

Results

Experiment 1

Repetitive pulse TMS significantly impaired performance on both the feature (d' decrease 0.8: $t_{(7)}$=2.27, p<0.05, Fig. 1a) and the conjunction tasks (d' decreased by 0.8: $t_{(7)}$=3.01, p<0.05, Fig. 1b).

Experiment 2

There was no effect of TMS on the feature search task when the first 100 ms after the onset of the visual array was left unstimulated ($t_{(5)}$=0.02, p>0.05, Fig. 1c). In the conjunction task rTMS reduced d' by 0.4 ($t_{(5)}$=3.07, p<0.05, Fig. 1d).

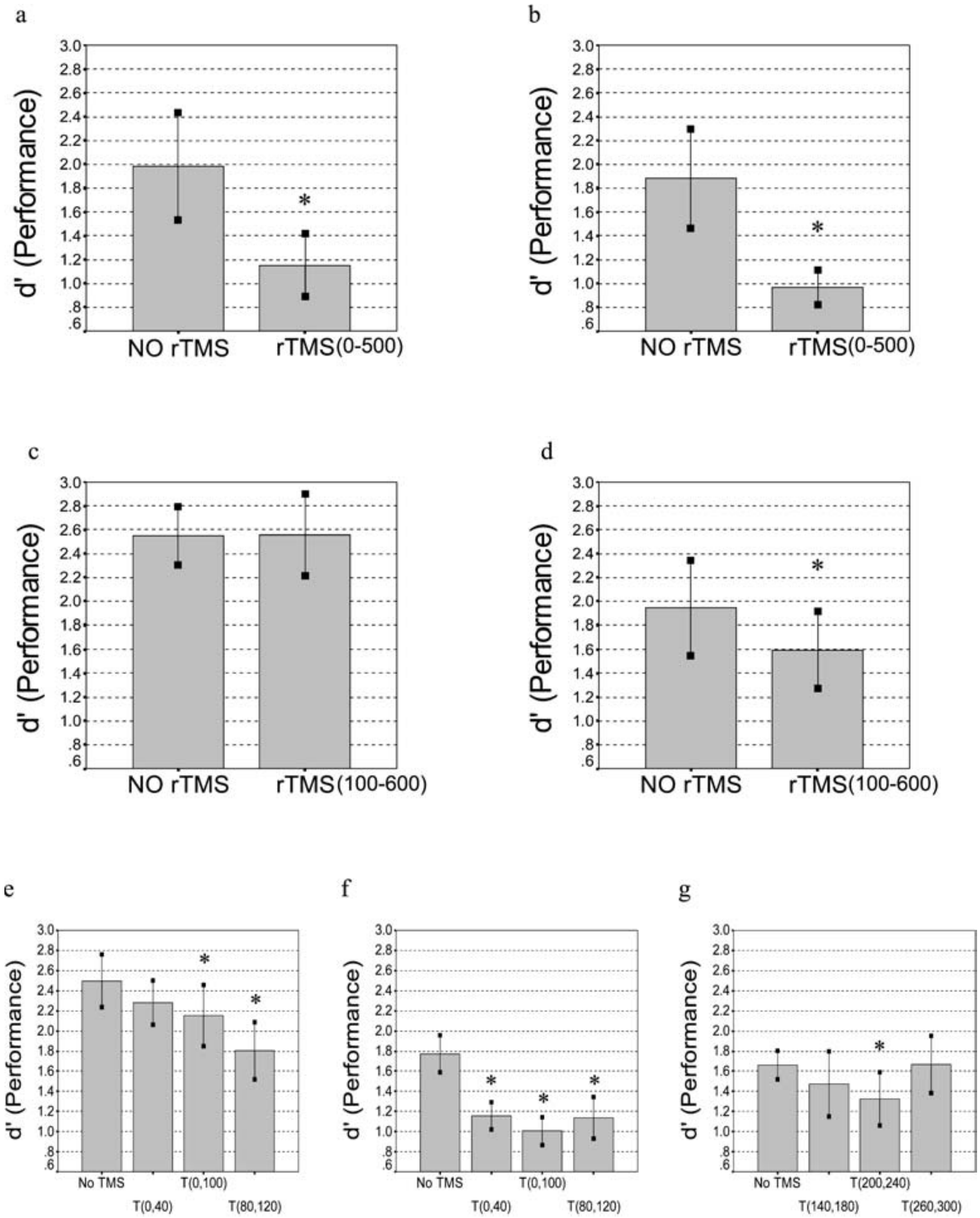

Fig. 1a–g The effects of TMS over V1 on feature and conjunction d' scores. Experiment 1: **a** feature search, **b** conjunction search. Experiment 2: **c** feature search, **d** conjunction search. Experiment 3: **e** feature search, **f, g** conjunction search

Experiment 3

In the feature task, dTMS produced a significant cost in d' ($F_{(3,5)}=6.26$, $p<0.01$, Fig. 1e) when applied at 0 and 100 ms ($t_{(5)}=2.85$, $p<0.05$) and at 80 and 120 ms ($t_{(5)}=6.05$, $p<0.01$) after array onset. In the conjunction task dTMS significantly impaired performance ($F_{(3,5)}=10.98$, $p<0.01$, Fig. 1f) when applied at any of the three early intervals: 0, 40 ($t_{(5)}=3.84$, $p<0.05$); 0, 100 ($t_{(5)}=5.09$, $p<0.01$); and 80, 120 ms ($t_{(5)}=14.42$, $p<0.01$) and one of the later intervals: 200, 240 ms ($t_{(5)}=1.72$, $p=0.053$, Fig. 1g).

Discussion

Interfering with V1 for 500 ms after visual stimulus onset shows V1 to be required in both visual feature and conjunction tasks. Delaying the onset of rTMS for 100 ms after visual stimulus onset shows that V1 is only required in the first 100 ms for feature search, but is important beyond the first 100 ms in conjunction search. Applying dTMS shows V1 to be involved in feature search in the later part of the first 100 ms and in the conjunction task throughout the first 100 ms and again between 200 and 240 ms.

These findings establish that V1 is necessary for normal performance in tasks that might be termed attentional (Somers et al. 1999; Brefczynski and DeYoe 1999; Nothdurft et al. 1999; Motter 1993; Tootell et al.

1998) and are consistent with the view that V1 receptive fields are influenced by configurations outside the classical receptive field (CRF). The data are also consistent with human event-related potential work which shows that back projections to V1 are functional earlier than has previously been thought (Foxe and Simpson 2002). The results therefore strongly support the reverse hierarchy theory (RHT) of visual perception (Ahissar and Hochstein 2000).

In the conjunction search task, TMS disrupted performance when applied in the first 120 ms after onset of the visual array (Fig. 1f). There are two consistent, and not mutually exclusive, explanations for this finding. When presented with a low signal to noise ratio visual array, as in a conjunction task, it takes time for visual analysis to extract the signal. This can occur either by repeated iterations within V1 itself or by repeated iterations of feedback and feed forward between V1 and higher areas (Lee et al. 1998; Foxe and Simpson 2002). The range of response latencies in monkey V1 is typically between 20 and 120 ms after the onset of simple visual stimulus (median latency 45–60 ms; see Bullier 2001), and in human cortex Foxe and Simpson (2002) have estimated a mean occipital onset of activity around 56 ms after stimulus onset. Both these estimates leave time for feedback to contribute to all but the very earliest of V1 responses between 20 and 40 ms (see Bullier 2001 for review).

The source of the back projections is not easy to identify, not due to a paucity of evidence but due to an abundance of candidates. Zipser et al. (1996), for example, noted that temporal stream activity preceded contextual modulation of V1 responses. Secondary visual neurones have response latencies that overlap greatly with the V1 response envelope and the latency of the feedback can be as fast as 10–15 ms (Pascual-Leone and Walsh 2001). Dorsal stream areas in general are activated before ventral stream areas, perhaps as early as 30 ms after stimulus onset (see Foxe and Simpson 2002), allowing time for multiple iterations between V1 and other areas. There is ample reason, then, to concur with Foxe and Simpson's claim that 'the time frame of "early" signal transmission is likely to be significantly more condensed' than is usually posited.

The existence of a later effect of dTMS in the conjunction task suggests two distinct roles for the back projection to V1. Following the gap between 120 and 200 ms, during which TMS does not interfere with performance, there is another period of V1 involvement which can be disrupted by dTMS at 200–240 ms (Fig. 1g). This later time window is consistent with a re-registration of extrastriate outputs which may need to be checked and localised by the detailed retinotopy of V1 CRFs (Ahissar and Hochstein 2000). Thus we propose early, multiple feedback/forward cycles in initial signal analysis and a second period of checking in V1 following "higher" level analysis. Another possibility is that the later V1 involvement reflects the subject's awareness of the detected target (see Cowey and Walsh 2000; Pascual-Leone and Walsh 2001).

The finding that TMS only disrupts feature detection when applied during the first 100 ms is consistent with the timing in other studies that have investigated the effects of TMS over V1 (e.g. Amassian et al. 1993; Corthout et al. 1999). The absence of an early effect in the 0, 40 condition suggests that for TMS to have an effect on a task that has a high signal content (simple features may be detected with redundancy in several parts of the visual system), pulses have to be administered at the critical time points. However, arrays with lower signal to noise values, such as conjunction arrays, can be disrupted by TMS at more times because *any* interference with processes involved in extracting a signal value will decrease performance. In other words, while the role of V1 is temporally narrow in easy feature extraction, it requires more time to process the noisier arrays.

The timings reported in this experiment targeted only a limited time window and used only a specific set of visual conditions. We are fully cognisant of the limitations this places on the role of V1 at other times and with different stimuli. For example, as Lamme and Roelfsma (2000) have observed, V1 activity that can be attributed to feedback processes can occur for up to 500 ms. Future studies will be required to extend the investigation of V1 to later time windows and to take account of the probability that the precise timing of V1 re-entrant effects will depend on the nature of the stimuli and the task at hand.

In summary, then, "early" is now "earlier" (Foxe and Simpson 2002); V1 is more than a distributor of simple attribute information (Bullier 2001; Ahissar and Hochstein 2000); what is fed forward may be determined by what is first fed back (e.g. early parietal activity; cf. Foxe and Simpson 2002); and extrastriate processing may be subject to a V1 veto (Ahissar and Hochstein 2000).

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